1. SNP BLUP (1)

Let $\mathbf{W}_1$ be on the scale of the allelic counts, but <u>centred</u>

$\mathbf{y} = \mathbf{1}\mu + \mathbf{W}_1\mathbf{b} + \mathbf{e}$ with $\mathbf{b} \sim MVN(0, \sigma_b^2\mathbf{I})$, then $\mathbf{u} = \mathbf{W}_1\mathbf{b}$ and $\hat{\mathbf{u}} = \mathbf{W}_1\hat{\mathbf{b}}$

2. Van Raden (1) is equivalent to SNP BLUP (1)

$\mathbf{y} = \mathbf{1}\mu + \mathbf{u} + \mathbf{e}$ with $\mathbf{u} \sim MVN(0, c^2\sigma_b^2(\mathbf{W}_1 / c)(\mathbf{W}_1^T / c))$, i.e. $MVN(0, \sigma_g^2\mathbf{G}_1)$

$\hat{\mathbf{u}}$ is obtained directly in this model but is fully equivalent to SNP BLUP (1).

One can test the assertion (but not prove) that $\hat{\mathbf{b}} = \mathbf{W}_1^T\mathbf{G}_1^{-1}\hat{\mathbf{u}} / c^2$ by $\hat{\mathbf{u}} = \mathbf{W}_1\hat{\mathbf{b}} = \mathbf{W}_1\mathbf{W}_1^T\mathbf{G}_1^{-1}\hat{\mathbf{u}} / c^2 = \mathbf{G}_1\mathbf{G}_1^{-1}\hat{\mathbf{u}} = \hat{\mathbf{u}}$.

So $\hat{\mathbf{b}} = \mathbf{W}_1^T\mathbf{G}_1^{-1}\hat{\mathbf{u}} / c^2$ is a solution for SNP effects on the scale of allele counts via Van Raden (1).

3. SNP BLUP (2)

Let $\mathbf{W}_2$ be on the scale of standard deviations, so $\mathbf{W}_2 = \mathbf{W}_1\mathbf{D}^{-1/2}$ where $\mathbf{D}$ is a <u>diagonal</u> matrix of variances for counts at each variance, assumed $2p_j(1-p_j)$ for locus j from HWE. This is standardising (or normalising) each covariate.

$\mathbf{y} = \mathbf{1}\mu + \mathbf{W}_2\boldsymbol{\beta} + \mathbf{e}$ with $\boldsymbol{\beta} \sim MVN(0, \sigma_\beta^2\mathbf{I})$

$\mathbf{u} = \mathbf{W}_2\boldsymbol{\beta}$ and $\hat{\mathbf{u}} = \mathbf{W}_2\hat{\boldsymbol{\beta}}$

4. Van Raden (2) is equivalent to SNP BLUP (2)

$\mathbf{y} = \mathbf{1}\mu + \mathbf{u} + \mathbf{e}$ with $\mathbf{u} \sim MVN(0, (m\sigma_\beta^2)(\mathbf{W}_2\mathbf{W}_2^T / m))$ or, as written, $\sigma_g^2\mathbf{G}_2$

$\hat{\mathbf{u}}$ is obtained directly from the calculation but is the fitted value from a fully equivalent model

One can test the assertion (but not prove) that $\hat{\boldsymbol{\beta}} = \mathbf{W}_2^T\mathbf{G}_2^{-1}\hat{\mathbf{u}} / m$ by $\hat{\mathbf{u}} = \mathbf{W}_2\hat{\mathbf{b}} = \mathbf{W}_2\mathbf{W}_2^T\mathbf{G}_2^{-1}\hat{\mathbf{u}} / m = \mathbf{G}_2\mathbf{G}_2^{-1}\hat{\mathbf{u}} = \hat{\mathbf{u}}$.

So $\hat{\boldsymbol{\beta}} = \mathbf{W}_2^T\mathbf{G}_2^{-1}\hat{\mathbf{u}} / m$ is solution for SNP effects on the standardised allele counts via Van Raden (2).

5. Comparison of SNP BLUP (1) and SNP BLUP (2)

If Van Raden (1) is re-written using the fact that $\mathbf{D}^{-1/2}\mathbf{D}^{1/2} = \mathbf{I}$ then $\mathbf{y} = \mathbf{1}\mu + (\mathbf{W}_1\mathbf{D}^{-1/2})(\mathbf{D}^{1/2}\mathbf{b}) + \mathbf{e}$ and $\mathbf{y} = \mathbf{1}m + \mathbf{W}_2(\mathbf{D}^{1/2}\mathbf{b}) + \mathbf{e}$ with $\mathbf{b} \sim MVN(0, \sigma_b^2\mathbf{I})$. This can be written $\mathbf{y} = \mathbf{1}\mu + \mathbf{W}_2\mathbf{b}^* + \mathbf{e}$ where $\mathbf{b}^* = \mathbf{D}^{1/2}\mathbf{b}$ with (co)variance matrix $E[\mathbf{D}^{1/2}\mathbf{b}(\mathbf{D}^{1/2}\mathbf{b})^T] = \mathbf{D}^{1/2}E[\mathbf{b}\mathbf{b}^T](\mathbf{D}^{1/2})^T = \sigma_b^2\mathbf{D}^{1/2}\mathbf{I}(\mathbf{D}^{1/2})^T = \sigma_b^2\mathbf{D}$, so this substitution results in a <u>different</u> model from Van Raden (2) because the distributional assumptions of $\mathbf{b}^*$ and $\boldsymbol{\beta}$ are different, even though both $\mathbf{b}^*$ and $\boldsymbol{\beta}$ are on the standardised scale, as $\sigma_b^2\mathbf{D} \neq \sigma_b^2\mathbf{I}$

To see the impact of the distributional assumptions either on the s.d. scale by comparing $\hat{\boldsymbol{\beta}}$ with $\mathbf{D}^{1/2}\hat{\mathbf{b}}$, or on the allele count (allelic substitution) scale by comparing $\mathbf{D}^{-1/2}\hat{\boldsymbol{\beta}}$ with $\hat{\mathbf{b}}$.

In the examples, the difference between Van Raden 1 and Van Raden 2 is particularly small as all markers have moderate MAF and so $\mathbf{D}$ looks similar to a scaled version of $\mathbf{I}$.